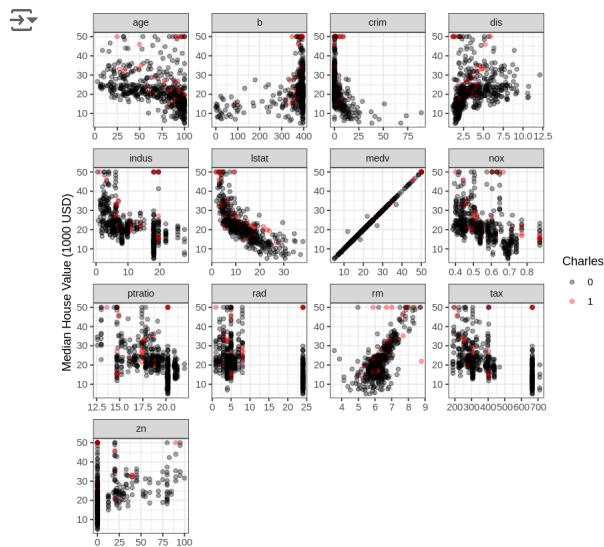


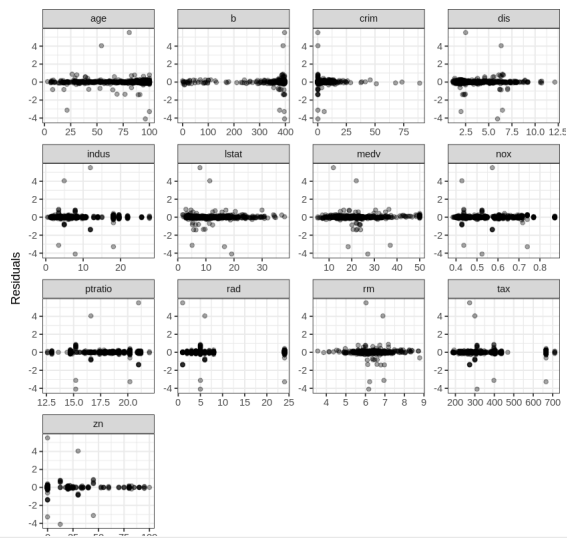
```
install.packages("mlbench")
library(mlbench)
data("BostonHousing2")
dat <- BostonHousing2
```

Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

```
library(dplyr)
library(ggplot2)
library(tidyr)
p <- dat %>%
  pivot_longer(-all_of(c("town", "tract", "lon", "lat", "cmedv", "chas"))) %>%
  ggplot(aes(x = value, y = cmedv, color = factor(chas))) +
  scale_color_manual(values = c("0" = "black", "1" = "red")) +
  geom_point(alpha = 0.35) +
  facet_wrap(~ name, scales = "free") +
  labs(x = "", y = "Median House Value (1000 USD)", color = "Charles") +
  theme_bw()
print(p)
```



```
fit <- lm(cmedv ~ ., data = dat)
residuals_df <- dat %>%
  mutate(residuals = residuals(fit)) %>%
  pivot_longer(-all_of(c("town", "tract", "lon", "lat", "cmedv", "chas", "residuals"))) %>%
  ggplot(aes(x = value, y = residuals)) +
  geom_point(alpha = 0.35) +
  facet_wrap(~ name, scales = "free") +
  labs(x = "", y = "Residuals") +
  theme_bw()
print(residuals_df)
```



```
summary(fit)
```



```
Call:
lm(formula = cmedv ~ ., data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.0966	-0.0506	0.0000	0.0509	5.5048

Coefficients: (5 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.765e+02	2.488e+02	1.111	0.2671
townAshland	-6.460e-01	1.418e+00	-0.455	0.6491
townBedford	4.788e-01	5.314e-01	0.901	0.3681
townBelmont	8.698e-02	2.826e-01	0.308	0.7584
townBeverly	3.773e+00	6.125e+00	0.616	0.5382
townBoston Allston-Brighton	7.661e+00	1.522e+01	0.503	0.6149
townBoston Back Bay	7.585e+00	1.482e+01	0.512	0.6090
townBoston Beacon Hill	7.669e+00	1.441e+01	0.532	0.5948
townBoston Charlestown	6.376e+00	1.352e+01	0.472	0.6375
townBoston Dorchester	5.117e+00	1.097e+01	0.466	0.6412
townBoston Downtown	5.925e+00	1.225e+01	0.484	0.6289
townBoston East Boston	6.411e+00	1.313e+01	0.488	0.6255
townBoston Forest Hills	4.661e+00	1.012e+01	0.460	0.6455
townBoston Hyde Park	4.150e+00	9.308e+00	0.446	0.6560
townBoston Mattapan	4.862e+00	1.057e+01	0.460	0.6458
townBoston North End	6.946e+00	1.398e+01	0.497	0.6195
townBoston Roxbury	5.474e+00	1.179e+01	0.464	0.6427
townBoston Savin Hill	5.101e+00	1.135e+01	0.449	0.6533
townBoston South Boston	5.861e+00	1.266e+01	0.463	0.6436
townBoston West Roxbury	4.420e+00	9.733e+00	0.454	0.6499
townBraintree	-2.040e+00	2.680e+00	-0.761	0.4469
townBrookline	-9.882e-01	1.951e+00	-0.506	0.6128
townBurlington	8.445e-01	1.151e+00	0.734	0.4635
townCambridge	1.199e-01	3.037e-01	0.395	0.6933
townCanton	-2.012e+00	2.514e+00	-0.800	0.4240
townChelsea	4.000e+00	8.406e+00	0.476	0.6345
townCohasset	-1.727e+00	2.869e+00	-0.602	0.5475
townConcord	5.213e-01	6.008e-01	0.868	0.3861
townDanvers	3.860e+00	6.361e+00	0.607	0.5443
townDedham	-1.564e+00	1.977e+00	-0.791	0.4293
townDover	-1.060e+00	2.149e+00	-0.493	0.6220
townDuxbury	-3.604e+00	6.340e+00	-0.568	0.5700
townEverett	2.167e-01	6.747e-01	0.321	0.7483
townFramingham	-5.089e-01	1.280e+00	-0.397	0.6913
townHamilton	4.225e+00	6.287e+00	0.672	0.5019
townHanover	-3.863e+00	6.194e+00	-0.624	0.5331
townHingham	-3.610e+00	6.134e+00	-0.589	0.5564
townHolbrook	-2.247e+00	2.760e+00	-0.814	0.4161
townHull	-3.596e+00	6.109e+00	-0.589	0.5564
townLexington	-3.481e-01	3.560e-01	-0.978	0.3287
townLincoln	5.309e-01	6.425e-01	0.826	0.4091
townLynn	3.452e+00	6.504e+00	0.531	0.5959
townLynnfield	3.742e+00	6.404e+00	0.584	0.5593
townMalden	2.697e-01	7.095e-01	0.380	0.7040
townManchester	4.156e+00	6.204e+00	0.670	0.5033
townMarblehead	3.889e+00	6.711e+00	0.580	0.5626
townMarshfield	-3.812e+00	6.297e+00	-0.605	0.5453
townMedfield	-1.487e+00	2.196e+00	-0.677	0.4989
townMedford	2.629e-01	7.655e-01	0.343	0.7315
townMelrose	4.822e-01	9.604e-01	0.502	0.6158
townMiddleton	4.083e+00	6.330e+00	0.645	0.5193
townMillis	-1.725e+00	2.260e+00	-0.763	0.4457
townMilton	-1.834e+00	2.561e+00	-0.716	0.4743
townNahant	3.383e+00	6.748e+00	0.501	0.6164
townNatick	7.230e-02	1.196e+00	0.060	0.9518
townNeedham	-1.323e+00	2.021e+00	-0.655	0.5131
townNewton	-4.911e-01	8.034e-01	-0.611	0.5414
townNorfolk	-1.742e+00	2.341e+00	-0.744	0.4572
townNorth Reading	1.274e+00	1.359e+00	0.937	0.3492
townNorwell	-3.751e+00	6.238e+00	-0.601	0.5480
townNorwood	-1.837e+00	2.433e+00	-0.755	0.4505
townPeabody	3.638e+00	6.363e+00	0.572	0.5678
townPembroke	-4.021e+00	6.380e+00	-0.630	0.5289
townQuincy	-1.892e+00	2.620e+00	-0.722	0.4706
townRandolph	-2.191e+00	2.719e+00	-0.806	0.4207
townReading	9.627e-01	1.101e+00	0.874	0.3826
townRevere	3.953e+00	7.994e+00	0.495	0.6212
townRockland	-3.888e+00	6.149e+00	-0.632	0.5275
townSalem	3.218e+00	6.632e+00	0.485	0.6278
townSargus	3.187e+00	6.404e+00	0.498	0.6190
townScituate	-3.623e+00	6.269e+00	-0.578	0.5636
townSharon	-2.016e+00	2.479e+00	-0.813	0.4165
townSherborn	-6.905e-01	1.437e+00	-0.480	0.6311
townSomerville	-3.832e-02	3.366e-01	-0.114	0.9094
townStoneham	5.487e-01	9.367e-01	0.586	0.5584
townSudbury	2.148e-01	7.343e-01	0.292	0.7701
townSwampscott	3.661e+00	6.726e+00	0.544	0.5866
townTopsfield	4.370e+00	6.301e+00	0.694	0.4883
townWakefield	7.531e-01	1.040e+00	0.724	0.4693
townWalpole	-1.835e+00	2.359e+00	-0.778	0.4370

townWaltham	-2.477e-01	5.947e-01	-0.417	0.6772
townWatertown	-3.407e-01	6.906e-01	-0.493	0.6221
townWayland	-9.609e-02	6.898e-01	-0.139	0.8893
townWellesley	-1.049e+00	2.070e+00	-0.507	0.6127
townWenham	4.211e+00	6.225e+00	0.676	0.4991
townWeston	9.250e-02	6.851e-01	0.135	0.8927
townWestwood	-1.781e+00	2.392e+00	-0.745	0.4570
townWeymouth	-2.070e+00	2.807e+00	-0.737	0.4614
townWilmington	1.212e+00	1.261e+00	0.961	0.3369
townWinchester	5.652e-01	8.556e-01	0.661	0.5092
townWinthrop	4.889e+00	7.579e+00	0.645	0.5193
townWoburn	6.977e-01	1.074e+00	0.650	0.5164
tract	2.160e-03	4.290e-03	0.504	0.6149
lon	4.796e-01	2.913e+00	0.165	0.8693
lat	-5.899e+00	3.480e+00	-1.695	0.0909
medv	9.739e-01	7.669e-03	126.994	<2e-16 ***
crim	5.110e-04	4.080e-03	0.125	0.9004
zn	NA	NA	NA	NA
indus	NA	NA	NA	NA
chas1	-2.179e-02	1.179e-01	-0.185	0.8534
nox	-1.123e+00	7.978e-01	-1.407	0.1601
rm	1.282e-01	6.901e-02	1.858	0.0639
age	-2.125e-03	1.922e-03	-1.106	0.2696
dis	-1.203e-01	8.444e-02	-1.425	0.1549
rad	NA	NA	NA	NA
tax	NA	NA	NA	NA
ptratio	NA	NA	NA	NA
b	2.156e-04	4.218e-04	0.511	0.6095
lstat	-5.848e-03	7.564e-03	-0.773	0.4399

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5099 on 402 degrees of freedom
Multiple R-squared: 0.9975, Adjusted R-squared: 0.9969
F-statistic: 1586 on 103 and 402 DF, p-value: < 2.2e-16

```

hatvals <- hatvalues(fit)
loo_cv_risk <- mean((residuals(fit) / (1 - hatvals))^2)
loo_cv_risk

```



```

dat <- dat %>%
  mutate(log_crim = log(crim + 1e-6),
         log_dis = log(dis + 1e-6),
         lstat_poly1 = poly(lstat, degree = 2, raw = TRUE)[, 1],
         lstat_poly2 = poly(lstat, degree = 2, raw = TRUE)[, 2])

```

```
dat <- dat %>%
  mutate(log_crim = log(crim + 1e-6),
         log_dis = log(dis + 1e-6),
         lstat_poly1 = poly(lstat, degree = 2, raw = TRUE)[, 1],
         lstat_poly2 = poly(lstat, degree = 2, raw = TRUE)[, 2],
         lstat_c = pmax(0, lstat + 0.025))

fit_extended <- lm(cmedv ~ crim + zn + indus + chas + nox + rm + age + dis + rad + tax + ptratio + b +
                  lstat + log_crim + log_dis + lstat_poly1 + lstat_poly2 + lstat_c, data = dat)

anova(fit, fit_extended)

hatvals_extended <- hatvalues(fit_extended)
loo_cv_risk_extended <- mean((residuals(fit_extended) / (1 - hatvals_extended))^2)
loo_cv_risk_extended
```

A anova: 2 x 6

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	402	104.502	NA	NA	NA	NA
2	489	8245.870	-87	-8141.368	359.9809	0

17.9328609117563

```
library(conformalInference)
funs <- lm.funs()

x.mat <- model.matrix(~ . + log_crim + log_dis + lstat_poly1 + lstat_poly2 + lstat_c, data = dat)
y <- dat$cmedv
new_data.mat <- model.matrix(~ . + log_crim + log_dis + lstat_poly1 + lstat_poly2 + lstat_c, data = new_data)

conf.pred.res <- conformal.pred(x = x.mat, y = y, x0 = new_data.mat,
                              alpha = 0.05, train.fun = funs$train, predict.fun = funs$predict)

conf.pred.res
```

Warning message in matrix(x0, ncol = p):
"data length [19] is not a sub-multiple or multiple of the number of columns [115]"
Error in chol.default(crossprod(x) + lambda[j] * diag(v)): the leading minor of order 2 is not positive
Traceback:

```
1. train.fun(x, y)
2. chol(crossprod(x) + lambda[j] * diag(v))
3. chol.default(crossprod(x) + lambda[j] * diag(v))
4. .handleSimpleError(function (cnd)
. {
.   watcher$capture_plot_and_output()
.   cnd <- sanitize_call(cnd)
.   watcher$push(cnd)
.   switch(on_error, continue = invokeRestart("eval_continue"),
.         stop = invokeRestart("eval_stop"), error = invokeRestart("eval_error",
.           cnd))
. }, "the leading minor of order 2 is not positive", base::quote(chol.default(crossprod(x) +
.   lambda[j] * diag(v))))
```